

Lishui Fan

✉ flscode@zju.edu.cn

📍 Hangzhou, China

🌐 lishuifan

🆔 0009-0006-4602-4296

Education

Zhejiang University, Computer Science and Technology

- College of Computer Science and Technology.
- Research focus: LLM-based code generation and software engineering.

Hangzhou, China

Sept 2023 – present

Nanjing University, Software Engineering

- State Key Laboratory for Novel Software Technology.

Nanjing, China

Sept 2020 – June 2023

Tianjin University, Software Engineering

- GPA: top 10%.

Tianjin, China

Sept 2016 – June 2020

Experience

Baidu ERNIE Bot Team, Research Intern

Working on large language model applications and code intelligence for ERNIE Bot.

China

May 2026 – present

3 months

ByteDance Douyin AI4SE, Research Intern

Worked on AI for software engineering and code intelligence problems in practical development settings.

China

Apr 2026 – May 2026

2 months

Taotian Future Life Lab, Research Intern

Conducted applied AI research for future-life and intelligent systems.

China

Apr 2025 – Aug 2025

5 months

Research

- **LLM-based Code Generation** — training, evaluation, and optimization methods for code LLMs, including fault-guided fine-tuning, keyword-enhanced prompting, and reasoning-process reward learning.
- **Code-Change Understanding** — systematic studies of LLM capabilities on code review generation, commit message generation, and just-in-time comment update.
- **Industry Collaboration** — core contributor to projects with Huawei and Tencent (CCF-Huawei Populus Grove Fund, Tencent Rhino-Bird special project), translating practical software engineering needs into model training, evaluation, and optimization problems.

Publications

ReCode: Reinforcing Code Generation with Reasoning-Process Rewards

A CCF-A paper on reasoning-process reward learning for code generation.

Lishui Fan, Yu Zhang, Mouxiang Chen, Zhongxin Liu

arxiv.org/abs/2508.05170

ZeroCoder: Can LLMs Improve Code Generation Without Ground-Truth Supervision?

A preprint on label-free co-evolution of code and test generation.

Lishui Fan, Mouxiang Chen, Tingwei Zhu, Kui Liu, Xin Xia, Shanping Li, Zhongxin Liu

arxiv.org/abs/2604.07864

Balancing Latency and Accuracy of Code Completion via Local-Cloud Model Cascading

A collaborative Tencent-related paper on local-cloud model cascading for code completion.

Hanzhen Lu, Lishui Fan, Jiachi Chen, Qiuyuan Chen, Zhao Wei, Zhongxin Liu

arxiv.org/abs/2603.05974

Exploring the Capabilities of LLMs for Code-Change-Related Tasks

A CCF-A journal article studying LLMs for code-change-related tasks.

Lishui Fan, Jiakun Liu, Zhongxin Liu, David Lo, Xin Xia, Shanping Li

doi.org/10.1145/3709358

FGit: Fault-Guided Fine-Tuning for Code Generation

A CCF-A conference paper on fault-guided fine-tuning for code generation.

Lishui Fan, Zhongxin Liu, Haoye Wang, Lingfeng Bao, Xin Xia, Shanping Li

doi.org/10.1109/ASE63991.2025.00114

SEK: Self-Explained Keywords Empower Large Language Models for Code Generation

A CCF-A paper on keyword-enhanced code generation with LLMs.

Lishui Fan, Mouxian Chen, Zhongxin Liu

aclanthology.org/2025.findings-acl.324

Projects

CCF-Huawei Populus Grove Fund Project

Core research contributor to an industry-academic project on software engineering intelligence.

- Code LLM training and evaluation.
- Translation of real software engineering needs into research problems.

Tencent Rhino-Bird Special Project

Core contributor to a Tencent collaboration on code intelligence and code completion.

- Related collaborative work accepted to FSE 2026.
- Focus on latency-accuracy trade-offs for code completion.

Huawei Industry Collaboration Project

Core research contributor to a Huawei industry collaboration project.

- Practical software engineering scenarios.
- Code LLM optimization and evaluation.

Skills

Research Areas

LLM-based code generation, software engineering, code-change understanding, fault-guided fine-tuning, reinforcement learning, coding agents.